Types of Mining

Underground mining

Surface mining

Types of Mining

- Traditional mining methods fall into two broad categories based on locale:

  1) ***Surface mining***

  2) ***Underground mining:*** is usually classified in three categories of methods: unsupported, supported, and caving.

Surface mining	Underground mining
Scoop ore off surface of earth or dig big holes and scoop: ➢ Disturbs large area (large amount environmental destruction). ➢ Produces large amounts of waste rocks (spoil). ➢ Relatively safe for miners (Safer working conditions and a better safety record than underground mining). ➢ Cheaper (Low operating costs). ➢ More efficient.	Use adits and shafts to reach deeply buried ores: ➢Dig a deep vertical shaft, blast underground tunnels to get mineral deposit, remove ore or coal and transport to surface ➢Less resource recovered, more dangerous and expensive ➢Disturbs less land and produces less waste: i) Disturbs much smaller surface area (Usually less environmental damage). ii) Waste rock (spoil) often left in mine. ➢Dangerous (hazardous for miners; collapse, explosions (natural gas), and lung disease) ➢Expensive (High operating costs). ➢Less efficient.

Surface and underground mining: what's the difference?

	Surface mining	Underground mining
What does it involve?	The predominant exploitation method worldwide. Refers to two main types of mining: open-cast and open-pit. Both open-cast and open-pit mining involves digging a large pit in the ground down from surface level. The main difference between the two is the direction of mining. In open-cast mining, the pit progresses across the landscape whereas an open-pit mine is deepened and widened but does not migrate laterally.	Does not require a surface pit but involves clearing rock space beneath the surface.
Comparative benefits?	Generally cheaper than underground mining, with higher productivity (i.e. a higher output rate of ore), lower operating costs, and safer working conditions.	Likely to need less than one hectare at each site because there is no surface pit. Only seven percent of rock tonnage extracted is waste. If the mine waste is replaced back into the mined-out tunnels ('backfilling'), there is also little need for surface dumping. Unfortunately this is not very common practice because of the additional costs involved. Given these reduced surface impacts, the impact on communities, forest and biodiversity in the locality is also reduced.
	Environmental and social impacts are much greater because the large amount of land required – some are up to a kilometer deep and several kilometers in circumference.	Without adequate attention paid to tunnel safety and the quality and strength of tunnel supports, mines can collapse. This can be dangerous both for the miners

	productivity (i.e. a higher output rate of ore), lower operating costs, and safer working conditions.	there is no surface pit. Only seven percent of rock tonnage extracted is waste. If the mine waste is replaced back into the mined-out tunnels ('backfilling'), there is also little need for surface dumping. Unfortunately this is not very common practice because of the additional costs involved. Given these reduced surface impacts, the impact on communities, forest and biodiversity in the locality is also reduced.
Comparative benefits?		
Problems?	Environmental and social impacts are much greater because the large amount of land required– some are upto a kilometer deep and several kilometers incircumference. Frequently results in deforestation, relocation of communities, and removal of artisanal miners. Open pit mining, and to a lesser extent open-castmining, creates a huge amount of rock waste compared to underground mining. Some analysts have estimated that73 percent of rock tonnage extracted by surface mining is waste, which requires a great deal of space for storageand	Without adequate attention paid to tunnel safety and the quality and strength of tunnel supports, mines can collapse. This can be dangerous both for the miners underground as well as human, plant and animal life on the surface. South Africa has one of the highest mining fatality rates in the world and of 220 deaths in 2007, more than half were caused by rock falls in underground mining.